

Nicholas Medland

ELECTRONICS ENGINEERING STUDENT

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nicholasmedland.com

Technical Skills

- **Programming Languages:** Java, C, C++, Python, VHDL, MATLAB, JavaScript, TypeScript
- **Embedded & Digital Systems:** Arduino, Raspberry Pi, finite-state machines, RISC-V
- **Design & Fabrication:** Autodesk Fusion 360, Shapr3D, FDM 3D printing, Bambu Studio

Freelance

Designer and Fabricator

Feb 2026 - Present

- Translated staff requirements and on-site measurements into functional prototypes and finished parts, incorporating client feedback to improve fit, durability, usability, and workflow.
- Delivered projects ranging from fitted hospitality organizers to a four-foot tall display trophy.

Projects

To-Do List Receipt Printer

Sep 2026

- Built an ESP32-S3 powered receipt printer system that converts tasks submitted through an iPhone Shortcut into printed tickets using a secure Cloudflare queue and custom ESC/POS firmware.

Casino Card Shuffler

Sep – Dec 2025

- Designed and 3D printed millimetre-precision components, then integrated an Arduino, motors, sensors, and IR remote control into an automated card shuffler and dispenser.

VHDL Elevator Controller

Nov – Dec 2025

- Developed and simulated a VHDL based elevator controller using a finite-state machine to coordinate floor requests, movement, door operation, and status outputs.

Professional Experience

Premium Server and Bartender

Aug 2023 - Present

Canucks Sports and Entertainment, Vancouver, BC

- Delivered high-end dining experiences by setting up and maintaining premium service standards.
- Provided discreet, first-class service to an elite clientele consistently elevating their experience.

Technician

Jun 2024 – Aug 2025

Lang's Venture Inc, Langley, BC

- Installed, diagnosed, and maintained lottery equipment at retail locations, resolving hardware and system faults to minimize operational downtime.

Election Equipment Officer

Sep– Nov 2024

Elections BC, Coquitlam, BC

- Programmed and tested voting equipment and office systems, supported deployment and troubleshooting, and helped maintain reliable operations serving 23,116 voters.

Education

Bachelor of Applied Science, Electronics Engineering
Simon Fraser University, Burnaby, BC

Expected August 2029